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A Natural User Interface for Realistic Tactile Perception of Object Surface Texture

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***Abstract*—Haptic feedback can increase the realism when interacting with virtual objects in a virtual environment. In this study, a natural user interface for providing realistic tactile feedback for object surface textures was developed. Users can wear the interface device in a virtual environment and naturally interact with virtual objects as they interact with physical objects in the physical environment to perceive object surface texture. A recording device composed of an accelerometer and a force sensor was designed to record the surface texture of an object. A simulation device composed of an accelerometer, a force sensor, and a piezo actuator was designed to reproduce the tactile feedback. A neural network was used to train an approximate function to transform the recorded data to simulation data. User test shows that users can perceive the simulated surface texture with at least 70% of correct rate.**

Keywords- haptic feedback; surface texture; neural network; natural user interface

I. INTRODUCTION

Prior research shows that haptic feedback improves grasping, manipulation, learning, and user response in a virtual environment [1]. There are two types of haptic feedback: kinesthetic haptic feedback and tactile haptic feedback. Kinesthetic haptic devices can create force (pull, push, weight, and displacement) sensation. Tactile haptic devices can create touch (surface texture, temperature, shape, and stiffness) sensation. In this research, tactile haptic devices for reproducing object surface textures were studied.

There are many different kinds of actuators for creating a haptic device. For example, Kyung et al. [2] used a DC motor and a piezo pin array to develop an integrated tactile display system which can provide kinesthetic force, pressure distribution, vibration and slip. Hayward and Cruz-Hernandez [3] developed a tactile feedback device composed of 64 closely aligned piezoelectric actuators coupled with thin film to create lateral skin stretch.

Stanley et al. [4] used a flat, deformable layer of hollow silicone cells filled with coffee grounds to create a tactile feedback device. This device can achieve both variable stiffness and deformable geometry via air pressure and a technique called particle jamming. Jansen et al. [5] used magnetic field to drive magnetorheological fluid to create a haptic feedback device, called MudPad. The electromagnet array below the device provided a magnetic field to accumulate or disperse the magnetorheological fluid. Goto and Takemura [6] used electric field to change the characteristics of electro-rheological fluid (ERF) at a specific position in the ERF chamber to provide tactile bump display.

Bau et al. [7] used electric field to create a device, called TeslaTouch. When users move a finger on the screen, they can feel subtle vibrations. Nakamura and Yamamoto [8] proposed a multi-finger electrostatic tactile feedback system on a visual display. The system could render different frictional stimuli to different fingers.

The above mentioned haptic devices were not wearable. Users needed to lay their hands on the devices to perceive surface textures. Therefore, the use of the haptic devices might not be as natural as interacting with physical objects in the real world for acquiring tactile sensations.

Hashimoto et al. [9] used large voice coil speakers to create pushing and pulling force. Users wore the device on their palms to receive haptic feedback through speaker's vibrations. Nishimura et al. [10] combined a voice coil actuator and two DC motors to develop a wearable haptic device which can provide vertical motion, vertical vibration, and horizontal skin stretch. Minamizawa et al. [11] used two DC motors and two belts to design a wearable device, called Gravity Grabber, to simulate the weight of a virtual object.

Although some researchers developed wearable devices, most tactile feedbacks were not realistic. Romano and Kuchenbecker [12] and Culbertson et al. [13] rendered realistic haptic feedback using a voice-coil actuator mounted on a stylus. The device could detect user's current speed and normal force and provide corresponding haptic feedback. Although the device could render realistic surface texture, users needed to hold a stylus to feel the surface texture indirectly. Therefore, the usage of the device was not as natural as normal human behavior of acquiring surface texture sensations.

II. METHOD

In this study, a wearable natural user interface for rendering realistic surface texture was developed. Users can directly acquire surface texture using normal human behavior. The system contains a recording device for

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recording physical surface texture, a processor for processing the recorded data, and a simulating device for rendering realistic surface texture tactile sensation.

A. Record Texture Data

The recording device included an MPU6050 3-axis accelerometer, an FSR402 force sensor, and an Arduino MEGA 2560 control board. The MPU6050 3-axis accelerometer was fixed on a hard fixture and bound with the FSR402 force sensor using a rubber finger sheath, as shown in Finger 1. Users wore the recording device and moved their fingers in different directions to record the surface textures, as shown in Figure 2. The sampling rate of the device was 1000 Hz.

In this research, normal compressive force (f_n) and 3-axis acceleration (a_x, a_y, and a_z) were recorded. The x- and y- velocities were computed from a_x and a_y. Since the z-axis (axis perpendicular to a surface) has the largest and dominating vibration energy when perceiving an object surface, in this research, the one-axis acceleration (z- axis) was used to reduce signal dimensions and computational complexity. The output of the recording device was v_x, v_y, a_z, and Δf.

Six different surface textures were tested: (a) strip pad, (b) plastic bump pad, (c) rubber bump pad, (d) breadboard, (e) 80-grit sandpaper, and (f) plastic smooth pad, as shown in Figure 3.

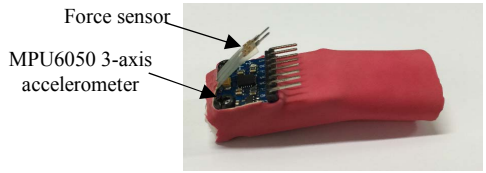


Figure 1. Finger part of the recording device

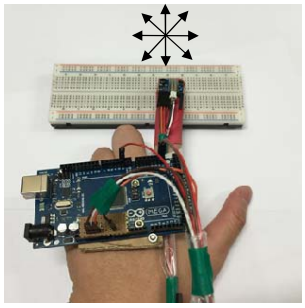


Figure 2. Recoding surface texture

B. Process Texture Data

In this study, a neural network was used to find the corresponding vibration accelerations. The input neurons included v_x, v_y, and Δf. The output layer was the corresponding recorded a_z. The neural network training would compute the connecting weights between the input layer and the hidden layer.

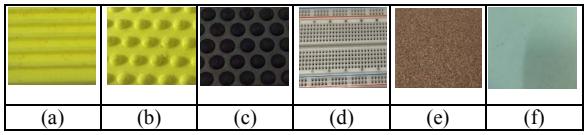


Figure 3. Six different test surface textures

TABLE I. NETWORK SETTING

Input layer neurons	Output neurons	Hidden layers	Hidden layer neurons	Active functions
3	1	1	10	logsig, purelin

Before the network training, band pass filter was used to filter out noises lower than 5 Hz caused by hand jittering and higher than 500Hz, frequency beyond human's tactile perception range. The setting of the neural network is shown in Table 1. Velocities v_x, v_y and Δf were set as the input data, and Z-axis acceleration a_z was set as the output data. The first one-third of data was used for training. The second one-third of data was used for validation. The last one-third of data was for testing.

The initial input weights, input bias weights, output weights, and output bias weights were randomly set between [-10, 10]. Equations (1) and (2) show the active function f_1 of the hidden layer (logsig) and the active function f_2 of the output layer (purelin), respectively. Equation (3) shows the transfer function of the hidden layer, where *net* is the hidden layer output, *input* is v_x,v_y, and Δf, w_{in} is the input weights, and $w_{b,in}$ is the input bias weights. Equation (4) shows the transfer function of the output layer, where *output* is the z-axis acceleration, w_{out} is the output weights, and $w_{b,out}$ is the output bias weights. The back propagation algorithm was used to train the neural network. The learning rate of the training process was set to 0.9. The steepest descent approximations and over-fitting control method were used to calculate the errors and modify the neuron weights. The training process stopped when it reached 2000 epochs or the total error was less than 10^{-9}.

$$f_1(n) = \frac{1}{1+e^{-n}} \tag{1}$$

$$f_2(n) = n \tag{2}$$

$$net = f_1\left(w_{in} \times input + w_{b,in}\right) \tag{3}$$

$$output = f_2\left(w_{out} \times net + w_{b,out}\right) \tag{4}$$

After the neural network was trained, the approximate transfer function was created. Using the approximate transfer function, acceleration corresponding to any velocities and forces can be found to simulate the acceleration caused by interacting with physical surface texture.

C. *Simulation*

The approximate transfer function was programmed into an Arduino MEGA2560, which was attached to the simulation device. The simulation device was consisted of a hard case for placing a finger, a 20 mm x 60 mm 96 v_{pp} piezo actuator, a 3-axis accelerometer, and a force sensor, as shown in Figure 4. The simulation device was designed to detect pressing force and at the same time to provide simulated texture vibration.

Users put their index finger into the simulation device and moved their finger with the simulation device to mimic the normal exploring behavior of rubbing a physical surface. The 3-axis accelerometer can detect instantaneous velocity and the force sensor can detect instantaneous compressive force. The data was sent to the Arduino MEGA2560 to calculate the corresponding acceleration. The acceleration was then amplified by an amplifier circuit to obtain the corresponding input voltage to drive the piezo actuator to create simulated vibration of the physical surface texture.

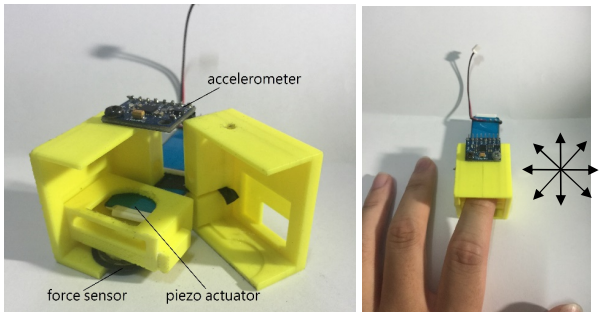


Figure 4. A simulation device

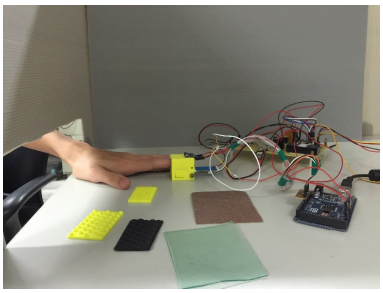
III. User Test And Results

Twenty participants aged between 20 and 35 took the user test. The user test included three parts: texture distinguish test, texture intensity test, and texture realism test. In order to prevent the test results affected by visual feedback, the participants were required to conduct the test without viewing the physical device setting, as shown in Figure 5.

A. *Texture Distinguish Test*

The purpose of the texture distinguish test was to test if the participants can distinguish a specific simulated surface texture of a corresponding physical surface texture. The intensities of the simulated surface textures were all set to be ± 45v, which was the maximum working voltage of the piezo actuator. Users wore the simulated device and moved their hands in different directions. For a given surface texture, based on their hand moving velocities and forces, a corresponding simulated texture vibration was created and output to the simulation device. Users then touched the six physical surface textures to choose the one which matched the simulated surface texture.

Figure 6 shows that participants can distinguish the strip pad much better than other surface textures. It might be the tactile sensations for the strip pad in different directions were obviously different. Participants can feel obvious different tactile sensations when they rub the strip pad in different directions.



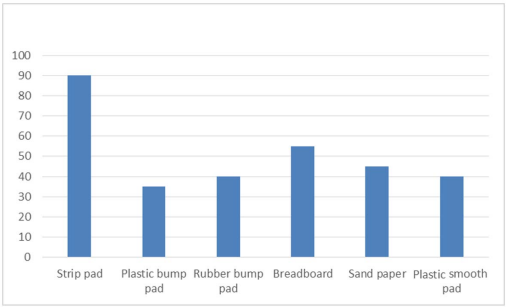


Figure 5. User test

Figure 6. Correct rate of the texture distinguish test

B. *Texture Intensity Test*

Since the recorded z-acceleration for different surface texture might be different, the driving voltage of the actuators for simulating different surface texture might be different. The purpose of the texture intensity test was to find the effects of the feedback intensity on the realism of the simulation.

Participants first touched a real texture, then they were given a corresponding simulated haptic feedback. The intensities of the feedbacks were controlled by 0.25 (low), 0.50 (medium-low), 0.75 (medium-high), and 1.00 (high) times of the maximum voltage (± 45v). After participants finished experiencing the simulated feedbacks, they chose a level which matched the physical surface texture the most.

Most participants felt that medium intensity feedback can simulate the strip pad better, high intensity feedback can

simulate the plastic bump pad better, medium-high intensity feedback can simulate the rubber bubble pad better, low intensity feedback can simulate the breadboard better, medium-low intensity feedback can simulate the sandpaper better, and low intensity feedback can simulate the plastic smooth pad better.

C. *Texture Realism Test*

The purpose of the texture realism test was to test if the participants can distinguish a simulated surface texture better with the corresponding intensity obtained from the texture intensity test. Similar to the texture distinguish test, but with different voltage intensity, participants experienced a simulated surface texture. Then, they touched the six physical surface textures and chose the one matched the given simulated surface texture.

Figure 7 shows that participants can distinguish the simulated plastic smooth pad and simulated strip pad with the corresponding physical surface textures quite well. But they distinguish the simulated rubber bump pad with the lowest correct rate. The reason might be because the rubber bump pad was soft and deformable. It was difficult to simulate soft and deformable properties by vibration. However, all of results had at least 70% of correct rate.

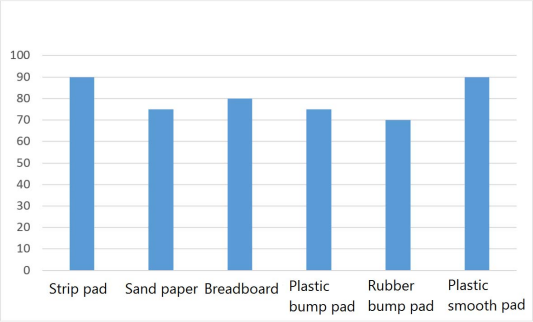


Figure 7. Correct rate for the texture realism test

IV. CONCLUSIONS

A lot of haptic feedback devices transmit haptic sensations to users using handheld tools, which were not intuitive or conform to human natural operation behaviors. This paper presents a method for recording the real world texture and rendering simulated tactile sensations using a natural user interface.

A recording device was design to record the physical texture. The recorded data was processed by an approximate function, which was trained through a neural network, to find input voltage information for the simulation device. The simulation device can simultaneously calculate finger's moving velocities and compression force, and provide corresponding vibration to simulate the physical texture tactile feedback. A user test was conducted, and results showed that users can correctly distinguish the simulated surface texture with at least 70% of correct rate.

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